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Design of Local Area Network

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1 Introduction

This project focuses on designing a scalable and cost-effective backbone network for the University of Moratuwa and the internal Local Area Network (LAN) of the Department of Electronic and Telecommunication Engineering (ENTC) building. The design considers important factors such as scalability, redundancy, bandwidth requirements and efficient IP addressing while following standard networking best practices.

The ENTC LAN and the proposed backbone network are simulated using **Cisco Packet Tracer** to verify connectivity, routing and overall network functionality. The Packet Tracer simulation files used in this project can be accessed through the following link:

Cisco Packet Tracer Simulation Files

2 Design Requirements and Assumptions

The design of the University of Moratuwa backbone network and the ENTC building LAN is guided by specific requirements and assumptions to ensure scalability, reliability, performance and cost-effectiveness. The key design considerations are outlined below.

2.1 Scalability Considerations

The network is designed to accommodate current users as well as future growth in the number of students, faculty and devices:

- Sufficient IP address space is allocated using IPv4 VLSM and IPv6 /64 prefixes to support all departments and shared facilities.(table 1)
- VLAN-based segmentation allows additional departments or units to be added without major reconfiguration.(table 2)
- The hybrid mesh-ring topology ensures that new nodes can be added easily while maintaining redundancy and connectivity.
- Access and distribution layer devices are selected with enough capacity to handle future bandwidth requirements.

2.2 Redundancy Requirements

High availability and fault tolerance are essential for a university network:

- Multiple paths between backbone nodes ensure that if one link or device fails, traffic can be rerouted automatically.
- Core and distribution switches are deployed with redundant uplinks to prevent single points of failure.
- Spanning Tree Protocol (STP) is implemented to prevent loops in the Layer 2 network.
- Dynamic routing (OSPF) ensures optimal path selection and automatic re-routing in case of link failures.

2.3 Cost Considerations

The design balances performance and reliability with cost efficiency:

- Enterprise-grade switches and access points are selected based on both capacity and budget constraints.
- Use of structured cabling and shared backbone infrastructure reduces overall deployment costs.
- Passive components, such as patch panels, racks and cables, are chosen for durability and cost-effectiveness.

3 University Backbone Network Design

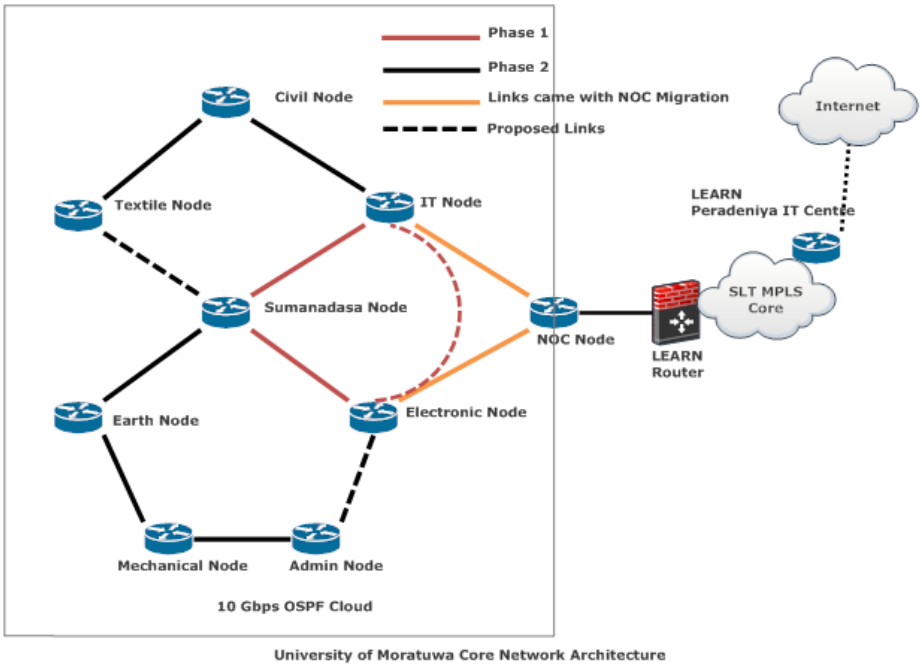


Figure 1: University of Moratuwa Core Network Architecture. Source: UoM CITeS

The university backbone network is designed to provide a high-speed, reliable and scalable communication infrastructure that interconnects all faculties, departments and other common areas within the University of Moratuwa. The backbone acts as the central data highway, enabling efficient data transfer between different parts of the campus network.

3.1 Design Approach

A hierarchical network design approach is used to ensure scalability, manageability and performance. The network is divided into three main layers:

- **Core Layer:** The core layer forms the backbone of the entire university network. It is responsible for fast and reliable data transport between different distribution layers. High-performance Layer 3 switches are used at this layer to provide high bandwidth and low latency communication.
- **Distribution Layer:** The distribution layer connects different faculties and buildings to the core layer. It performs routing, policy enforcement, inter-VLAN routing and traffic control. This layer improves network organization and security.
- **Access Layer:** The access layer connects end devices such as computers, printers and wireless access points to the network. It provides VLAN-based segmentation and ensures secure user access.

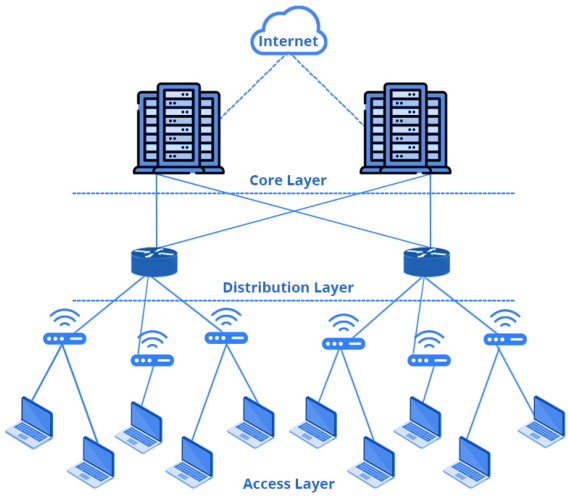


Figure 2: 3-Tier Architecture

This layered approach improves network scalability, simplifies troubleshooting and allows future expansion without major redesign.

3.2 Network Topology

The university backbone network follows a **hybrid topology** combining features of both **mesh** and **ring** topologies. This design is selected to achieve high reliability, redundancy and efficient data routing across the campus.

- Core network devices are interconnected in a ring-like structure, ensuring continuous connectivity even if a single link fails.
- In addition to the ring structure, extra links are introduced between key nodes, forming partial mesh connections.
- As a result, each node in the backbone has **multiple paths** to reach other nodes in the network.

This hybrid approach provides the following advantages:

- **Redundancy:** Multiple paths ensure that failure of a single link or device does not disrupt network communication.
- **Fault Tolerance:** Traffic can be rerouted dynamically through alternative paths using routing protocols.
- **Improved Performance:** Load can be distributed across multiple links, reducing congestion.
- **Scalability:** New nodes can be added with additional links without affecting the overall structure.

Therefore, the hybrid mesh-ring topology provides a balance between performance, cost and reliability, making it suitable for a large campus network.

3.3 IP Addressing Scheme

3.3.1 IPv4 Addressing Plan

A structured IPv4 addressing scheme is used based on the private IP range `10.0.0.0/8`. The address space is divided using Variable Length Subnet Masking to efficiently allocate IP addresses based on the number of users in each department.

- Large faculties (e.g., Architecture, Information Technology) are assigned larger subnets such as /20 or /21.

Department / Unit	Intake (Approx)	4 Batches + Staff	IP Allocation	No. of Bits
Faculty of Engineering (FoE)				
CE	115	575	1022	10
EE	100	500	1022	10
ENTC	115	575	1022	10
ME	100	500	1022	10
MT	50	250	510	9
CH	80	400	1022	10
CS	200	1000	2046	11
ER	80	400	1022	10
EM	80	400	1022	10
TMLE	98	490	1022	10
TEXT	80	400	1022	10
Other Faculties				
FoIT	400	2000	4094	12
ARCHI	650	3250	4094	12
Business	100	500	1022	10
MED	100	500	1022	10
Common				
Library	–	1000	2046	11
CENTRA	–	1000	2046	11
WALA	–	500	1022	10
MATH DEPT	–	500	1022	10
ADMIN	–	1000	1022	10

Table 1: Estimated Users and IP Address Allocation for University LAN Design

- Medium sized departments are assigned /22 subnets.
- Smaller units are assigned /23 subnets.

Each department is mapped to a unique VLAN, which improves network segmentation, security and traffic management.

3.3.2 IPv6 Addressing Plan

An IPv6 addressing scheme is implemented to support future scalability and modern networking requirements. The network uses a global unicast IPv6 prefix assigned to the network.

- **IPv6 Prefix:** 2401:dd00:548::/48

IPv6 addresses are assigned using **Stateless Address Autoconfiguration (SLAAC)**. In this method, devices automatically generate their own IPv6 addresses by combining the network prefix with an interface identifier.

- **Address Assignment Method:** SLAAC

The Layer 3 switch sends **Router Advertisements (RA)** to inform devices about the network prefix and default gateway. Using this information, hosts can configure their own IPv6 addresses without the need for a DHCP server.

- **Default Gateway:** Link-local address of VLAN interface (learned via RA)

The use of SLAAC provides several advantages:

- Automatic IP address configuration without manual setup
- No requirement for a DHCP server
- Simplified network management
- Efficient support for a large number of devices

A dual-stack approach (IPv4 and IPv6) is implemented to ensure compatibility with existing IPv4-based systems while enabling IPv6 communication.

VLAN	Department	Host Bits	Usable Hosts	Network	Start IP	End IP
10	ARCHI	12	4094	10.0.0.0/20	10.0.0.1	10.0.15.254
20	FoIT	11	2046	10.0.16.0/21	10.0.16.1	10.0.23.254
30	CS	11	2046	10.0.24.0/21	10.0.24.1	10.0.31.254
40	LIB	11	2046	10.0.32.0/21	10.0.32.1	10.0.39.254
60	ADMIN	10	1022	10.0.40.0/22	10.0.40.1	10.0.43.254
80	ENTC	10	1022	10.0.44.0/22	10.0.44.1	10.0.47.254
50	CENTRA	11	2046	10.0.48.0/21	10.0.48.1	10.0.55.254
70	CE	10	1022	10.0.56.0/22	10.0.56.1	10.0.59.254
90	EE	10	1022	10.0.60.0/22	10.0.60.1	10.0.63.254
100	ME	10	1022	10.0.64.0/22	10.0.64.1	10.0.67.254
110	BUSI	10	1022	10.0.68.0/22	10.0.68.1	10.0.71.254
120	MED	10	1022	10.0.72.0/22	10.0.72.1	10.0.75.254
130	WALA	10	1022	10.0.76.0/22	10.0.76.1	10.0.79.254
140	MATH DEPT	10	1022	10.0.80.0/22	10.0.80.1	10.0.83.254
150	CH	10	1022	10.0.84.0/22	10.0.84.1	10.0.87.254
160	ER	10	1022	10.0.88.0/22	10.0.88.1	10.0.91.254
170	EM	10	1022	10.0.92.0/22	10.0.92.1	10.0.95.254
180	TEXT	10	1022	10.0.96.0/22	10.0.96.1	10.0.99.254
190	TMLE	10	1022	10.0.100.0/22	10.0.100.1	10.0.103.254
200	MT	9	510	10.0.104.0/23	10.0.104.1	10.0.105.254

Table 2: IP Allocation and VLAN Assignment for University LAN

3.4 Routing Strategy

The routing strategy for the University of Moratuwa backbone and the ENTC building LAN is designed to ensure efficient, scalable and reliable data delivery across all departments.

- **Backbone Routing (Cisco Catalyst 3650 Switches):** Open Shortest Path First (OSPF) is used as the dynamic routing protocol between distribution and core switches. OSPF provides:
  - Fast convergence and adaptive path selection
  - Scalability across multiple VLANs and departments
  - Support for multiple paths in the hybrid mesh-ring topology
- **Internal Department Routing:** Within each department (such as ENTC), routing is handled using **Shortest Path Tree (SPT)**. This ensures:

- Optimal routing between devices inside the department network
- Low latency communication between PCs, servers and local resources
- Predictable traffic paths that minimize loops and congestion

Summary: Using OSPF for inter-department communication and SPT for intra-department traffic ensures a balance between scalability, efficiency and low complexity within the network.

3.5 Backbone Network Simulation

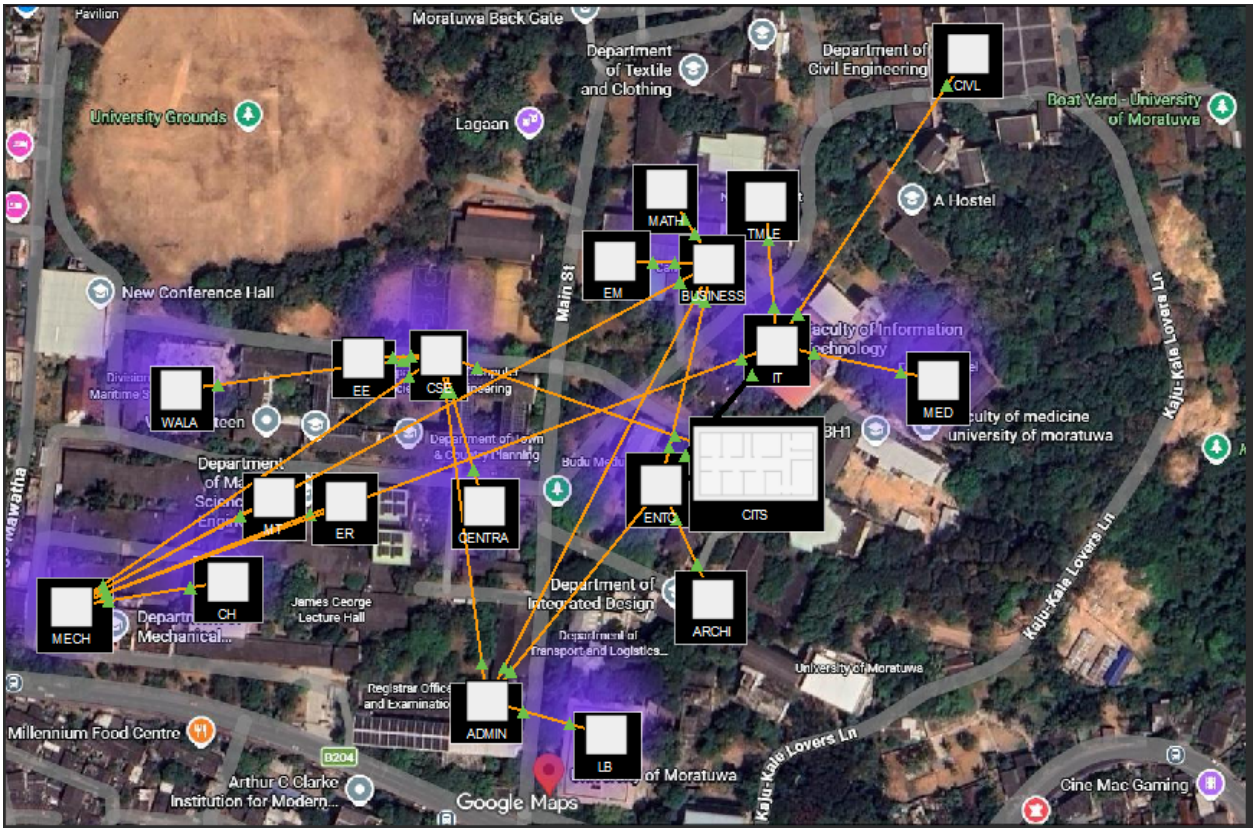


Figure 3: UoM physical Topology

3.5.1 DHCP (IPv4 and IPv6) Configuration

```
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::250:FFF:FEB7:C34A
   IPv6 Address....................: 2401:DD00:548:10:250:FFF:FEB7:C34A
   IPv4 Address....................: 10.0.0.3
   Subnet Mask.....................: 255.255.240.0
   Default Gateway.................: FE80::2E0:A3FF:FE3B:4801
                                     10.0.0.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

Figure 4: IP configuration of PC1 in ARCHI faculty

```
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::202:4AFF:FEA4:83D3
   IPv6 Address....................: 2401:DD00:548:60:202:4AFF:FEA4:83D3
   IPv4 Address....................: 10.0.40.2
   Subnet Mask.....................: 255.255.252.0
   Default Gateway.................: FE80::290:2BFF:FEAA:B802
                                     10.0.40.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

Figure 5: IP configuration of PC7 in LIBRARY

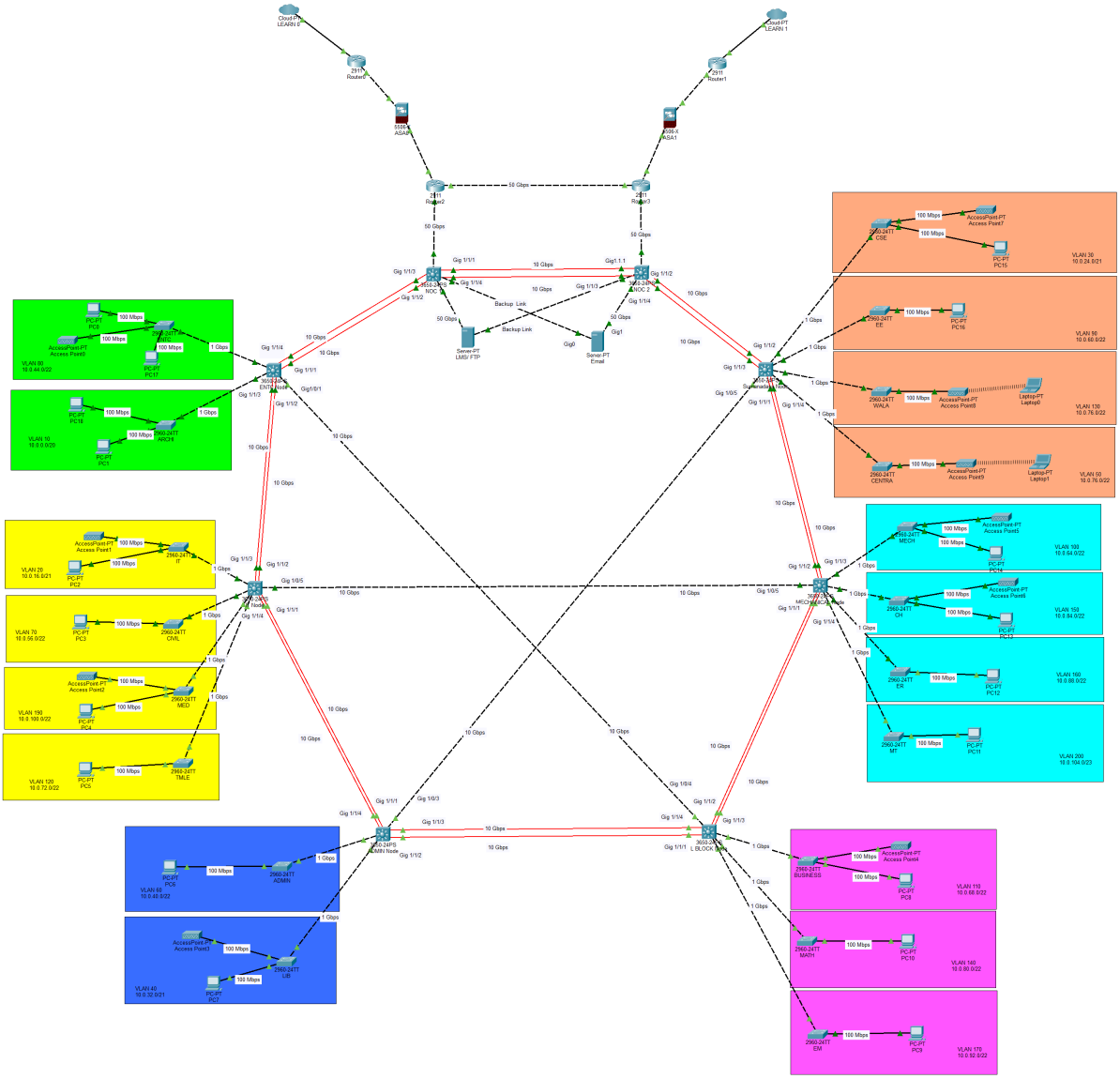


Figure 6: UoM Backbone Logical Topology

The above figure illustrates the logical backbone topology of the University of Moratuwa network implemented in the simulation. It shows how core, distribution and access layer devices are interconnected to provide end-to-end communication across different faculties and units. The topology follows a hybrid mesh-ring structure, ensuring multiple redundant paths for data transmission. This design improves reliability, supports efficient routing using OSPF and enables scalable expansion of the campus network.

3.5.2 Connectivity Verification

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC8	PC9	ICMP		0.000	N	0	(edit)	(delete)
	Successful	Laptop1	PC15	ICMP		0.011	N	1	(edit)	(delete)
	Successful	PC6	PC8	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC1	Laptop0	ICMP		0.000	N	3	(edit)	(delete)

Figure 7: Simple PDU transmissions between inter and intra nodes (All are successful)

IPv4 Ping Tests:

```
C:\>ping 10.0.44.3

Pinging 10.0.44.3 with 32 bytes of data:

Reply from 10.0.44.3: bytes=32 time<1ms TTL=125
Reply from 10.0.44.3: bytes=32 time<1ms TTL=125
Reply from 10.0.44.3: bytes=32 time<1ms TTL=125
Reply from 10.0.44.3: bytes=32 time<1ms TTL=125

Ping statistics for 10.0.44.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 8: Ping tests between ENTC & ADMIN

Traceroute Tests

```
C:\>tracert 10.0.44.3

Tracing route to 10.0.44.3 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      10.0.40.1
  2   4 ms      0 ms      0 ms      10.255.255.5
  3   0 ms      0 ms      0 ms      10.10.255.2
  4   0 ms      10 ms     0 ms      10.0.44.3

Trace complete.
```

Figure 9: Traceroute results for PC6 (Library) and PC17 (ENTC)

Application Layer Testing

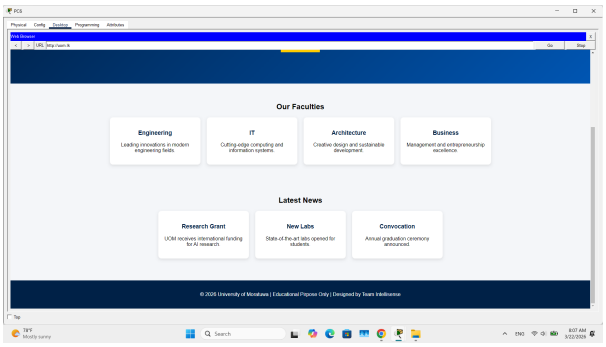


Figure 10: PC accessing a university HTML webpage (`uom.lk`) in the simulation, confirming successful network connectivity and web server operation.

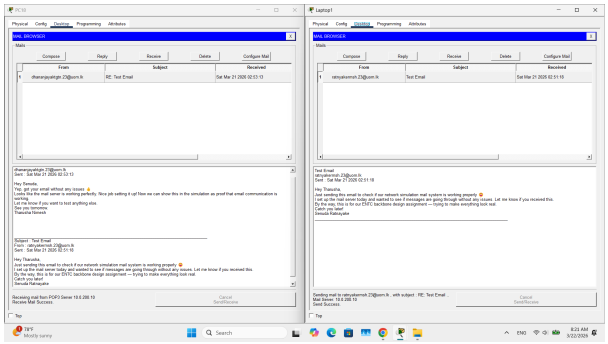


Figure 11: Email communication between `ratnayakermsh.23@uom.lk` and `dhananjayaktgtn.23@uom.lk` demonstrating successful mail server operation.

3.5.3 OSPF Verification

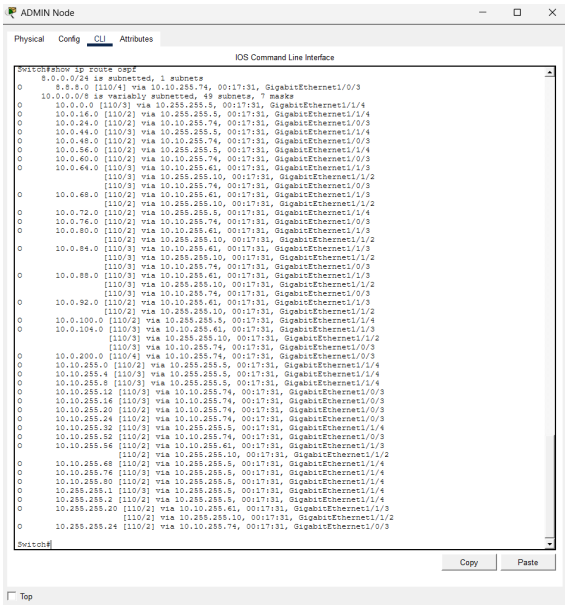


Figure 12: OSPF verification

4 ENTC Local Area Network Design

4.1 Overview of Existing LAN

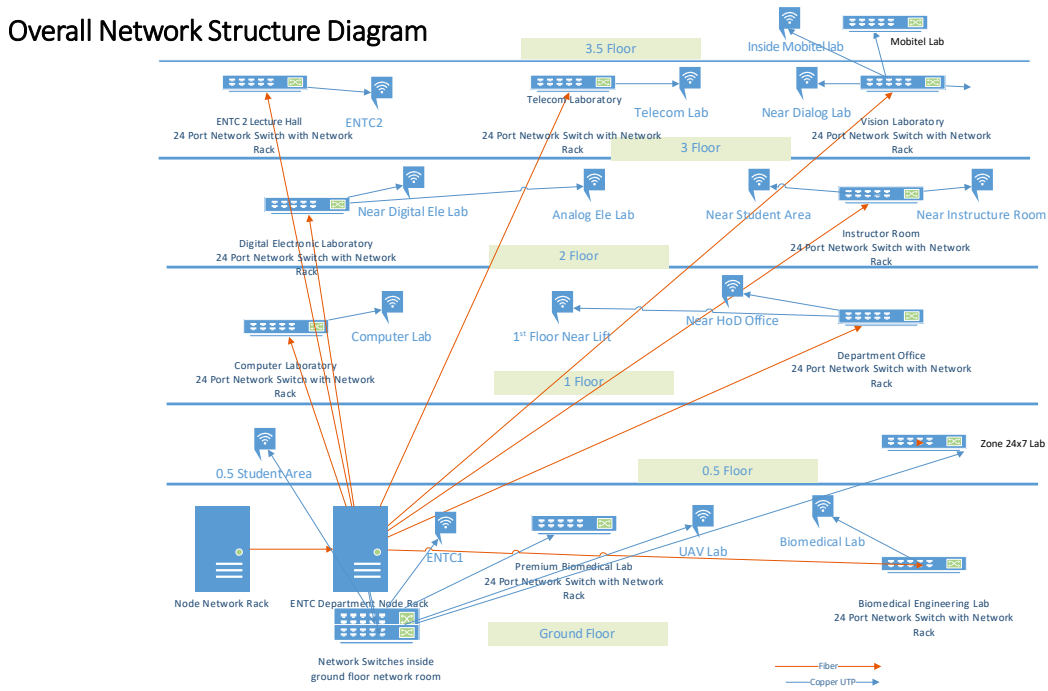


Figure 13: ENTC Overall Network Structure Diagram. **ACK:** Mr. Damith Kandage.

4.2 ENTC Network Design

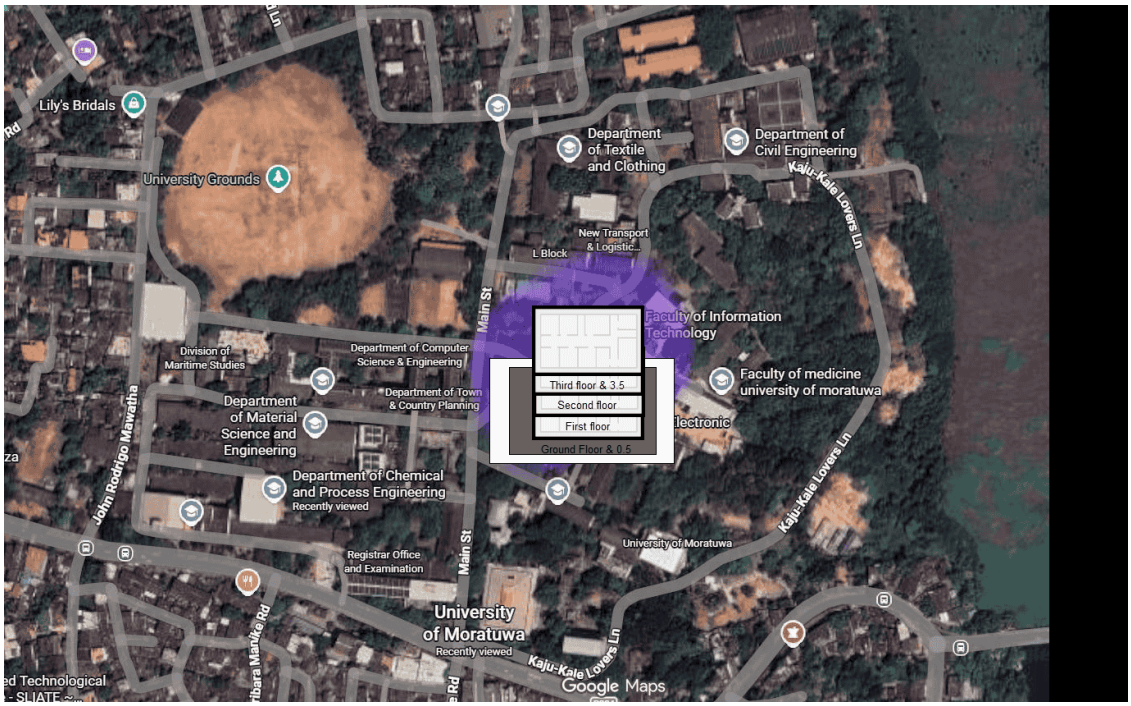


Figure 14: Physical network topology of ENTC

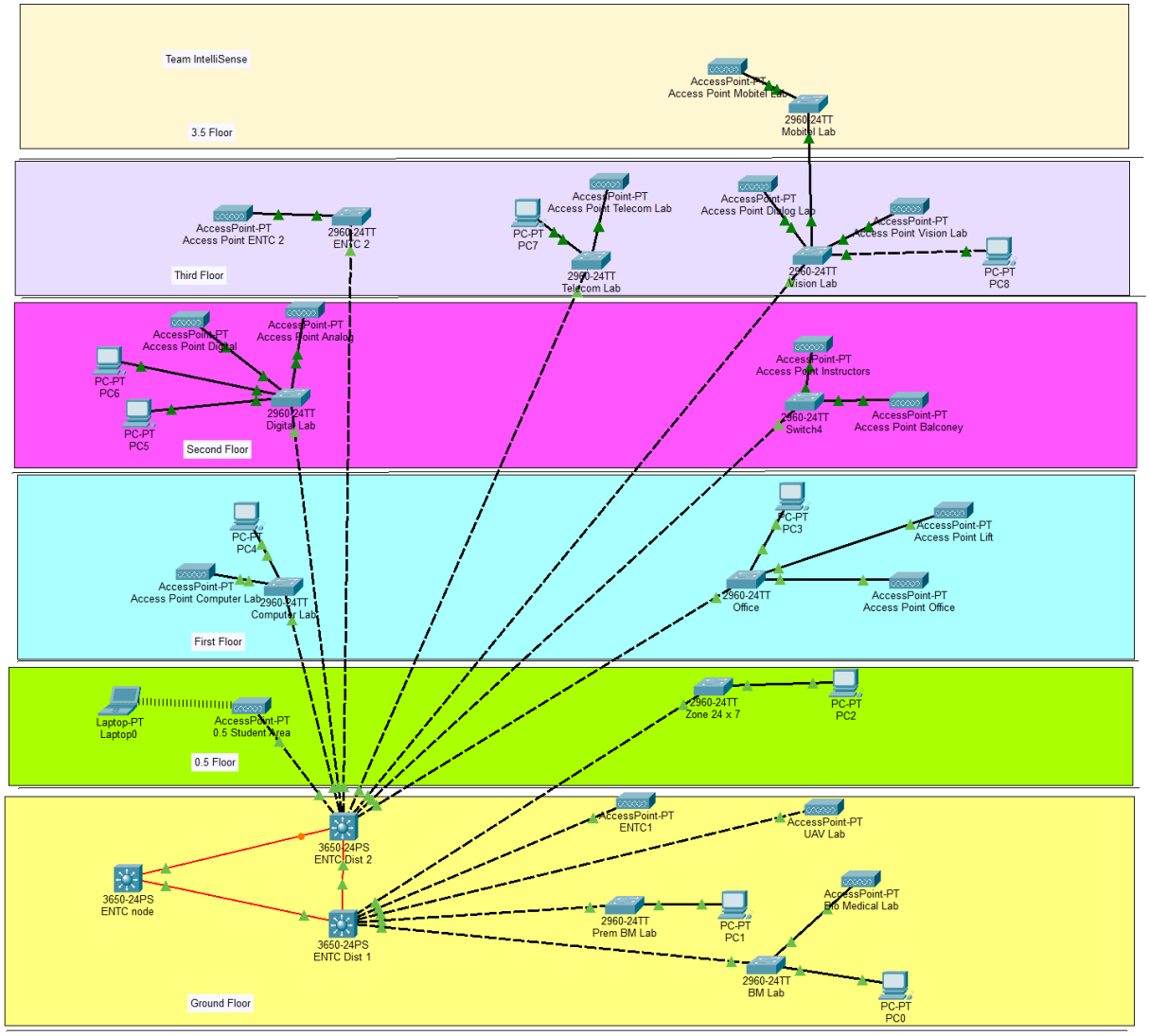


Figure 15: ENTC Logical Network Architecture. Due to the use of Spanning Tree Protocol (STP), some redundant links (e.g., between ENTC node and ENTC Dist 2) are in the blocking state.

4.3 ENTC Network Simulation Results

The ENTC building network is simulated using Cisco Packet Tracer. The network includes multiple PCs, servers, switches, routers and uses DHCP for automatic IP assignment

4.3.1 DHCP (IPv4 and IPv6) Configuration

- **IPv4 DHCP Subnet:** 10.0.44.0/22
- **Subnet Mask:** 255.255.252.0
- **IP Range for Hosts:** 10.0.44.2 – 10.0.47.254
- **Default Gateway:** 10.0.44.1 (Layer 3 switch VLAN interface)
- **DNS Server:** 8.8.8.8
- **IPv6 Prefix:** 2401:dd00:548:1::/64
- **IPv6 Address Assignment:** Stateless Address Autoconfiguration (SLAAC)
- **IPv6 Default Gateway:** FE80::202:17FF:FECD:98C5 (VLAN interface link-local address)

```
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2E0:B0FF:FE48:9549
   IPv6 Address....................: 2401:DD00:548:1:2E0:B0FF:FE48:9549
   IPv4 Address....................: 10.0.44.3
   Subnet Mask.....................: 255.255.252.0
   Default Gateway.................: FE80::202:17FF:FECD:98C5
                                     10.0.44.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

Figure 16: IP configuration of PC0

```
Wireless0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2D0:97FF:FE61:90B9
   IPv6 Address....................: 2401:DD00:548:1:2D0:97FF:FE61:90B9
   IPv4 Address....................: 10.0.44.7
   Subnet Mask.....................: 255.255.252.0
   Default Gateway.................: FE80::202:17FF:FECD:98C5
                                     10.0.44.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

Figure 17: IP configuration of Laptop0

4.3.2 Connectivity Verification

Connectivity verification is performed to ensure that all devices in the ENTC network can communicate with each other properly. Different network testing tools such as ping, traceroute and Packet Tracer simulation are used to confirm that the configured IP addresses, routing and switching operations work correctly.

IPv4 Ping Tests:

```
C:\>ping 10.0.44.7

Pinging 10.0.44.7 with 32 bytes of data:

Reply from 10.0.44.7: bytes=32 time=26ms TTL=128
Reply from 10.0.44.7: bytes=32 time=19ms TTL=128
Reply from 10.0.44.7: bytes=32 time=14ms TTL=128
Reply from 10.0.44.7: bytes=32 time=18ms TTL=128

Ping statistics for 10.0.44.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 26ms, Average = 19ms
```

Figure 18: Ping results with PC0 and Laptop0

Ping tests are used to check whether two devices can communicate through the network. The successful ping replies between PC0 and Laptop0 indicate that the devices received valid IP addresses from DHCP and that the network configuration allows proper data transmission between hosts.

Traceroute Tests

```
C:\>tracert 10.0.44.3

Tracing route to 10.0.44.3 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      10.0.44.3

Trace complete.
```

Figure 19: Traceroute results for PC4 and PC0

Traceroute is used to identify the path that packets take from a source device to a destination device. The traceroute results confirm that packets travel through the expected router interfaces and reach the destination successfully, verifying correct routing configuration.

Wireless Connection

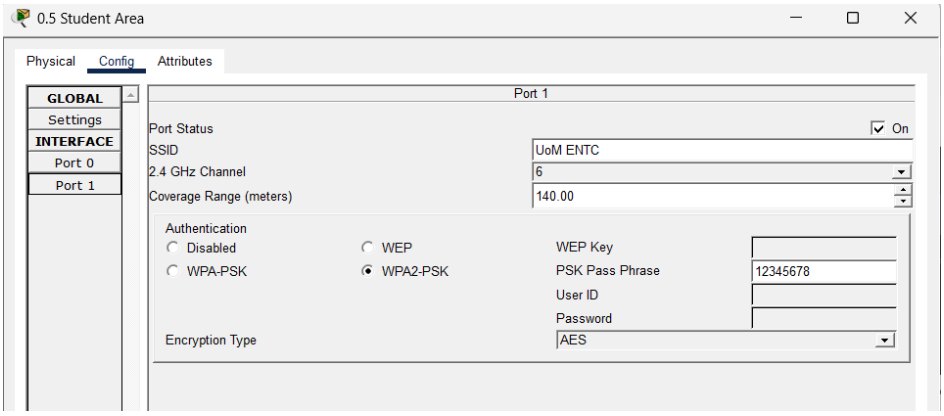


Figure 20: Wireless connection with internet connectivity

The wireless connection test verifies that wireless devices can connect to the network through the access point. Successful connectivity shows that the wireless configuration allows devices to obtain IP addresses and access network resources, including the internet.

Simple PDU transmissions

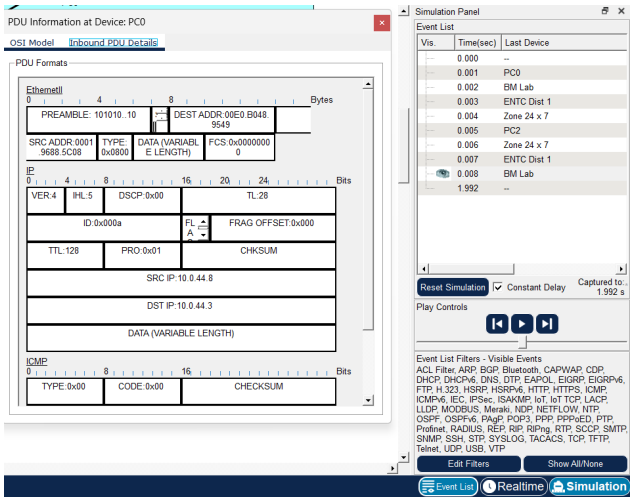
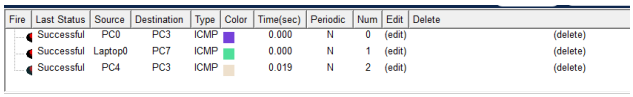


Figure 21: Simple PDU transmission between PC0 and PC1

Simple PDU transmission in Cisco Packet Tracer is used to simulate packet delivery between devices. The successful transmission between PC0 and PC1 confirms that the network switches and routers correctly forward packets between hosts.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	Laptop0	PC7	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC4	PC3	ICMP		0.019	N	2	(edit)	(delete)

Figure 22: Simple PDU transmissions simulation results

5 Component Selection and Justification

The network infrastructure for the University of Moratuwa backbone network and the ENTC building LAN consists of both **active components** and **passive components**. Active components perform network

processing tasks such as switching, routing and security management, while passive components provide the physical infrastructure required for network connectivity.

5.1 Active Network Components

5.1.1 Cisco Catalyst 3650 24-Port Switch

The Cisco Catalyst 3650 switches are used as **distribution/core layer switches** within the university backbone.

Justification

- Supports both Layer 2 and Layer 3 switching functions.
- Enables inter-VLAN routing within the building network.
- High switching capacity suitable for aggregating multiple access switches.
- Provides advanced security features and Quality of Service (QoS).
- Designed for enterprise campus networks.

Figure 23: Cisco Catalyst 3650 24-Port Switch

5.1.2 Cisco Catalyst 2960-L 24-Port Gigabit Ethernet Switch

The Cisco Catalyst 2960-L switches are used as **access layer switches** in the buildings network to connect end-user devices such as computers, printers and wireless access points.

Justification

- Provides 24 Gigabit Ethernet ports for connecting multiple user devices.
- Supports VLAN configuration for network segmentation and improved security.
- Reliable Layer 2 switching platform widely used in enterprise and campus networks.
- Energy efficient and cost-effective for large-scale LAN deployments.
- Suitable for connecting end devices at the access layer.

Figure 24: Cisco Catalyst 2960-L 24-Port Gigabit Ethernet Switch

5.1.3 Cisco Catalyst 9105AX Indoor Access Point

The Cisco Catalyst 9105AX access points provide wireless connectivity for students, staff and mobile devices.

Justification

- Supports Wi-Fi 6 (802.11ax) technology for higher speed and efficiency.
- Supports a large number of simultaneous wireless connections.
- Improves network performance in high-density environments.

- Suitable for lecture halls, laboratories and office environments.

Figure 25: Cisco Catalyst 9105AX Indoor Access Point

5.1.4 Cisco 2911 Router

The Cisco 2911 router is used for routing between the building network and the campus backbone network.

Justification

- Provides multiple Gigabit Ethernet interfaces for connecting to different network segments.
- Supports dynamic routing protocols such as OSPF and EIGRP.
- Suitable for medium-sized enterprise and campus networks.
- Provides reliable and scalable routing capabilities.

Figure 26: Cisco 2911 Router

5.1.5 Cisco ASA 5500-X Firewall

The Cisco ASA 5500-X firewall is used to secure the network by controlling incoming and outgoing traffic.

Justification

- Provides advanced network security and threat protection.
- Supports firewall filtering, VPN services and intrusion prevention.
- Protects the campus network from unauthorized access.
- Suitable for enterprise-level network security.

Figure 27: Cisco ASA 5500-X Firewall

5.1.6 APIC-SERVER-M1-RF

The APIC server is used as part of the network management and automation system.

Justification

- Provides centralized network management and automation.
- Supports Software Defined Networking (SDN) architectures.
- Improves network scalability and operational efficiency.
- Enables monitoring and configuration of large campus networks.

Figure 28: APIC-SERVER-M1-RF

5.1.7 Cisco ENCS 5408 Cloud Server

The Cisco ENCS 5408 is used as a network edge computing platform for hosting virtual network services.

Justification

- Supports virtualization of network services such as routing and firewall functions.
- High-performance computing platform with multi-core processors.
- Provides scalability and flexibility for future network services.
- Suitable for modern cloud-enabled campus networks.

Figure 29: Cisco ENCS 5408 Cloud Server

5.2 Passive Network Components

5.2.1 Copper Ethernet Crossover Cables

Copper crossover cables are used for connecting networking devices directly when required.

Justification

- Provides reliable short-distance connectivity.
- Cost-effective solution for LAN device connections.
- Easy to install and maintain.
- Suitable for connecting switches, routers and other network devices.

Figure 30: Copper Ethernet Crossover Cables

5.2.2 Single Mode Fibre Optic Cable

Single mode fibre optic cables are used for backbone connections between buildings in the campus network.

Justification

- Supports long-distance communication with minimal signal loss.
- Provides very high bandwidth capacity.
- Suitable for campus backbone networks connecting multiple buildings.
- Future-proof solution for high-speed data transmission.

5.2.3 Multimode Fibre Optic Cable

Multimode fibre cables are used for shorter fibre connections within buildings or between nearby network equipment.

Justification

- Cost-effective solution for short-distance high-speed fibre communication.
- Suitable for connections between switches within network racks.
- Provides high bandwidth for internal network communication.

Figure 31: Fibre optic cables

5.2.4 Network Racks

Network racks are used to organize and mount networking equipment such as switches, routers and patch panels.

Justification

- Provides structured installation of network devices.
- Improves cable management and equipment organization.
- Protects network hardware and allows efficient cooling.
- Facilitates easier maintenance and troubleshooting.

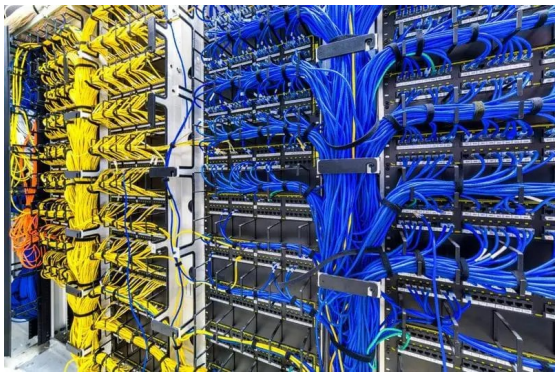


Figure 32: Network Racks

5.2.5 Patch Panels

Patch panels are used to terminate and organize network cables in the structured cabling system.

Justification

- Simplifies cable management and network organization.
- Allows easy reconfiguration of network connections.
- Reduces wear on switch ports by providing a fixed termination point.
- Commonly used in structured cabling systems in enterprise networks.

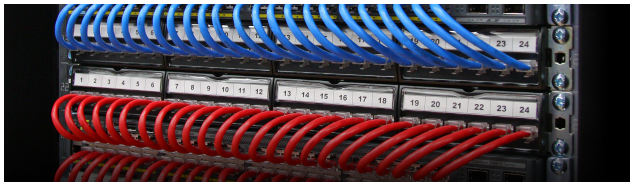


Figure 33: Patch Panels

Note: *All images in this section are sourced from publicly available internet resources.*

5.3 Backbone Network Device Features and Specifications

5.3.1 Cisco Catalyst 3650 (Backbone Switch)

- **Ports:** 24 x Gigabit Ethernet ports
- **Uplinks:** High-speed SFP uplink ports
- **Layer Support:** Layer 2 and Layer 3
- **Routing:** OSPF, EIGRP supported
- **DHCP Functionality:** The switch operates as an internal DHCP server, dynamically assigning IP addresses to VLANs without the need for a separate DHCP server
- **QoS:** Traffic prioritization
- **Security:** ACLs, port security, DHCP snooping
- **Scalability:** Stackable architecture

5.3.2 Cisco 2911 Router (Backbone Router)

- **Interfaces:** Multiple Gigabit Ethernet ports
- **Routing Protocols:** OSPF
- **Security:** VPN and firewall capabilities
- **Performance:** Suitable for medium-scale networks
- **Modularity:** Expandable with modules

5.3.3 Cisco ASA 5500-X Firewall

- **Firewall Throughput:** High-speed filtering
- **VPN:** Site-to-site and remote access
- **Security:** Intrusion prevention and access control
- **High Availability:** Failover supported

6 Bill of Quantities

6.1 Bill of Quantities (BoQ) for UoM Backbone Network

Name	Unit Price	Quantity	Sub Total
Cisco Catalyst 3650 24 Port	$3,372.00	8	$26,976.00
3-Metre Singlemode Fibre Optic Cable	$11.00	5000 m	$18,333.33
1-Metre Multimode Fibre Optic Cable	$22.00	500 m	$11,000.00
Cisco 2911 Router w/3 GE	$1,567.00	2	$3,134.00
APIC-SERVER-M1-RF	$8,666.25	2	$17,332.50
Cisco ENCS 5408 (8-core Intel, 16G DRAM) Cloud	$4,793.89	2	$9,587.78
Cisco ASA 5500-X Firewall	$851.00	2	$1,702.00
Racks	$259.99	2	$519.98
Other (Network sockets, Patch cords)	–	–	$500.00
		Total	$89,085.59

Table 3: Bill of Quantities (BoQ) for University Backbone Network

6.2 Bill of Quantities (BoQ) for ENTC Network

Name	Unit Price	Quantity	Sub Total
Catalyst 2960L 24 Port GigE	$1,063.00	11	$11,693.00
Cisco Catalyst 3650 24 Port	$3,372.00	2	$6,744.00
Cisco Catalyst 9105AX Indoor Access Point	$371.00	16	$5,936.00
Copper Cross Over (25 ft)	$9.98	5000 ft	$1,996.00
3-Metre Single mode Fibre Optic Cable	$11.00	1000 m	$3,666.67
Racks	$259.99	4	$1,039.96
Patch Panels	$34.98	4	$139.92
Other (Network sockets, Patch cords)	–	–	$200.00
		Total	$31,415.55

Table 4: Bill of Quantities (BoQ) for ENTC Network

References

[1] Router-Switch.com, "Global ICT Hardware Supplier." Available: https://www.router-switch.com/

[2] Cisco Systems, "Cisco Networking Products Data Sheets (Catalyst 2960-L Series Switches, Catalyst 3650 Series Switches, Catalyst 9105AX Access Points, Cisco 2900 Series Integrated Services Routers and Cisco ASA 5500-X Series Firewalls)." Available: https://www.cisco.com